

AMPC POS & Inventory Management System

End-User UAT Findings — UI/UX Test Session

Tester (as end-user): **Cashier & Admin walk-through** · Environment: **localhost:5173 (dev)** · Build: **main @ latest** · Date: **2026-09-21**

Scope. This session emulates the everyday operator — a cooperative cashier and a store admin — clicking through the login, dashboard, POS, inventory, forecasting, finance, and user-management screens as they exist in the running system. Findings are reported from a **usability** lens (clarity, safety of the click, discoverability), not a QA lens (unit-test coverage or code hygiene).

Method. Puppeteer + system Chrome drove 14 captures across both desktop (1440×900) and mobile (390×844) viewports. Screenshots are attached inline as evidence. See docs/uat/screens/ for source PNGs.

Result. 40 issues found — **3 Critical, 14 High, 16 Medium, 7 Low**. Nothing here blocks a demo, but several items would be caught quickly if the cooperative's real staff started using the app. Fixes are suggested per row.

Summary

Severity	Count	Themes
CRITICAL	3	Credential leak on login, redirect-without-message on protected route, currency inconsistency across screens.
HIGH	14	Ambiguous KPI labels, missing empty-state guidance, non-obvious click targets in POS, accidental-logout risk, no product images, alarming "Email alert" buttons, discount-modal password without show/hide.
MEDIUM	16	Inline vs modal Add Product, always-editable Min Stock cell, small-text diagnostics, missing tooltips, mixed button styles, sidebar branding, forecast-legend distinguishability.
LOW	7	Labels, tooltips, avatar menu, empty-state illustrations, keyboard shortcuts, cursor-pointer misuse.

Screen 1 — Login

Two captures: default landing state and the response to bad credentials.

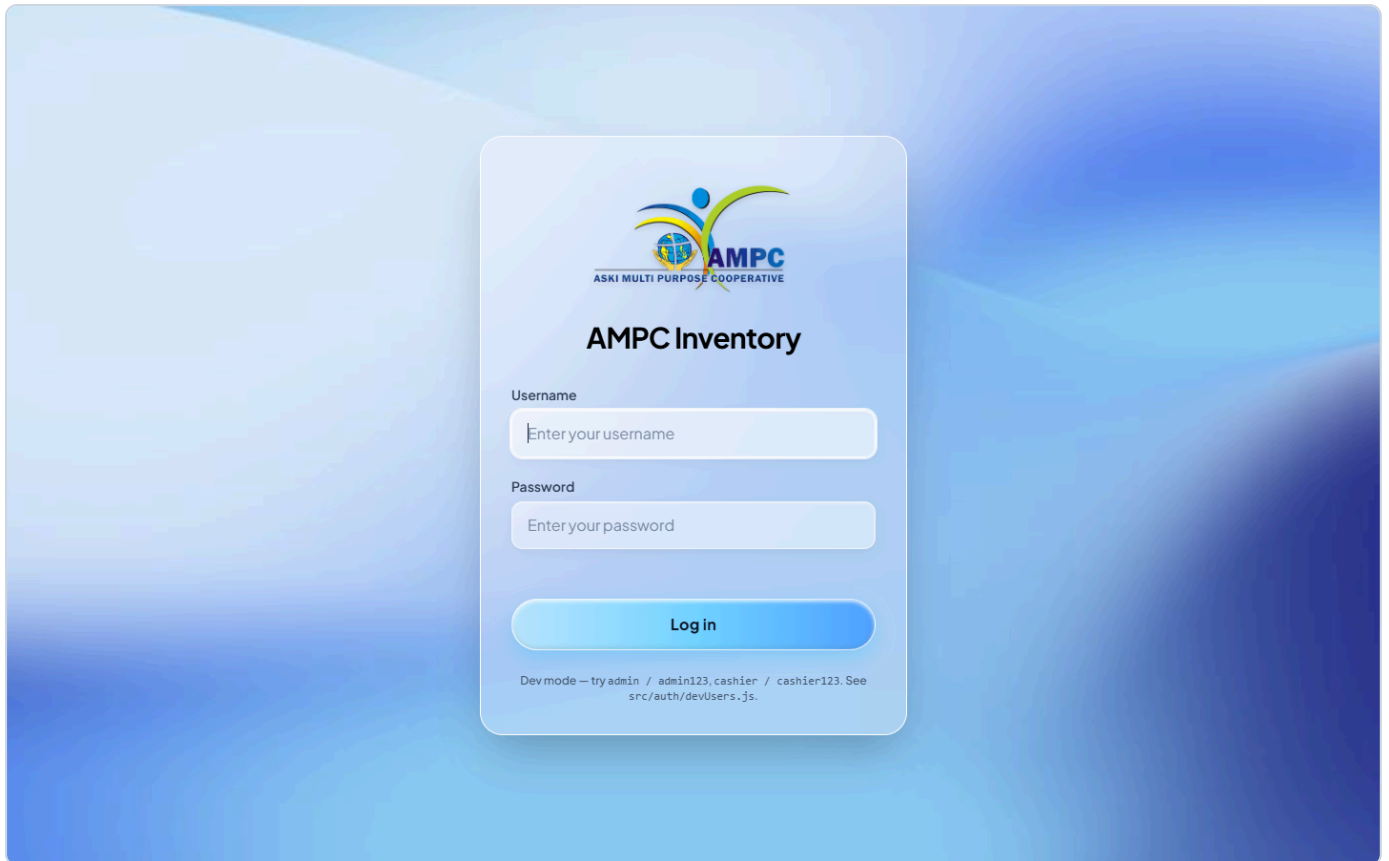


Figure 1a — Login page as an end user first sees it.

UAT-01 **CRITICAL** Login page exposes dev credentials and a source-file path publicly

Under the "Log in" button the page prints "Dev mode — try admin / admin123, cashier / cashier123. See src/auth/devUsers.js." This is visible to any visitor of the deployed URL, so it (a) leaks working credentials for two roles, (b) advertises "dev mode" to an outsider, and (c) reveals an internal filename. A real end user has no reason to see any of this.

Fix: gate the hint behind `import.meta.env.DEV` so it only renders on `npm run dev`, not on `npm run build/preview`. If it must appear in staging, put it behind a keyboard shortcut instead of on-screen text.

UAT-02 **HIGH** Password field has no show/hide toggle

Cashiers on shift with adjacent screens can't verify what they typed. Every other password field in the app (Admin Verification, Supervisor Discount) already has an eye icon — login is inconsistent.

Fix: reuse the existing `<PasswordInput />` pattern from the `AdminAuthModal`.

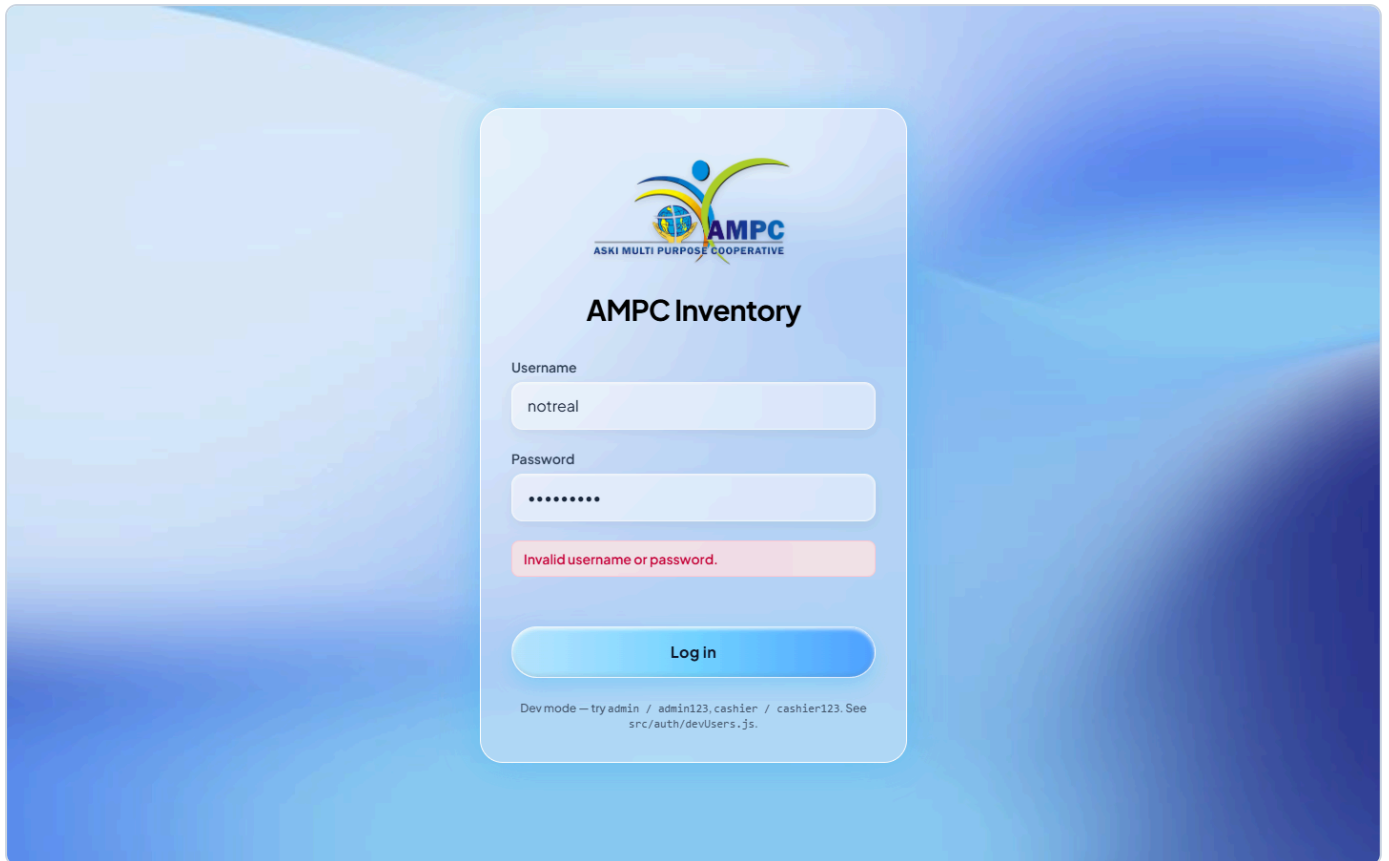


Figure 1b — Error message after typing wrong credentials.

UAT-03 **MEDIUM** Error message is generic and offers no next action

"Invalid username or password." tells the user nothing about which field failed, whether they're locked out, or how to recover a forgotten password. There is no "Forgot password?" link anywhere.

Fix: add a "Forgot password?" link that opens a contact-your-admin flow, and after 5 failed attempts show a "Locked — contact admin" state with an obvious next step.

UAT-04 **MEDIUM** The "AMPC Inventory" brand name is inaccurate for the app's scope

The system is a POS **and** an inventory manager. Calling it "AMPC Inventory" on the login card undersells it to a cashier who thinks they've opened the wrong app.

Fix: rename to "AMPC POS & Inventory" (matches the repo/README).

Screen 2 — Admin Dashboard

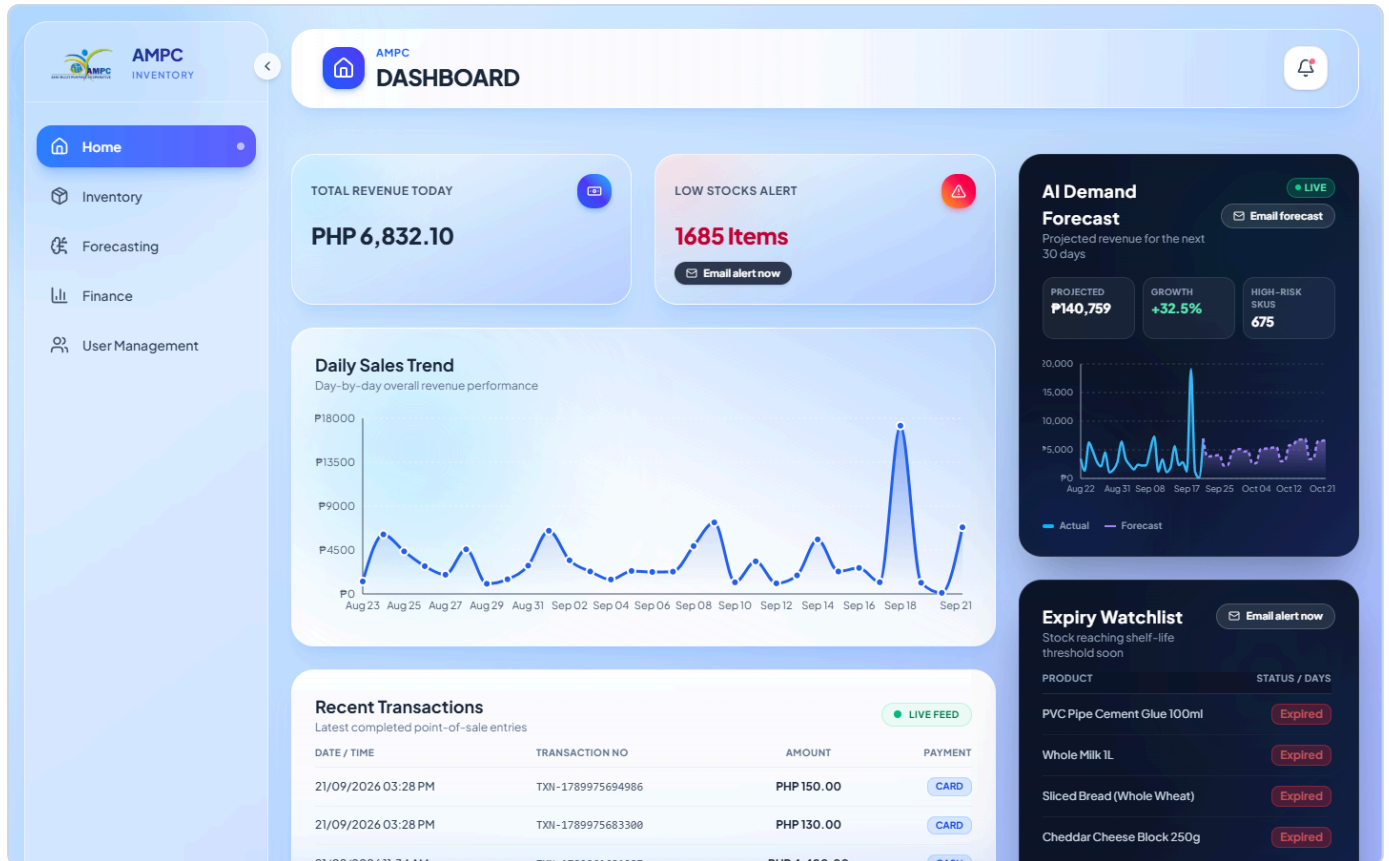


Figure 2 — Landing dashboard immediately after admin login.

UAT-05 **CRITICAL** Currency label swings between "PHP" and "₱" on the same screen

Total Revenue Today reads PHP 6,832.10, but the **AI Demand Forecast** tile just below it reads ₱140,759, and the Recent Transactions rows switch between PHP 150.00 and — on other screens — ₱150.00. It looks like two systems glued together.

Fix: single `formatPeso(n)` helper using ₱, applied everywhere. Add an ESLint rule to catch bare "PHP " prefixes.

UAT-06 **HIGH** "Total Revenue Today" doesn't state the date

In a seeded demo where dates span months, the admin can't tell what "today" means. This will be a support ticket the moment the system goes to a real coop that closes at 6 pm and the number never matches the register tape.

Fix: subtitle the card with "as of 21 Sep 2026, 03:35 PM" and refresh it on the same cadence as the socket.

UAT-07 **HIGH** AI Demand Forecast tile heading wraps awkwardly

"AI Demand" appears on one line and "Forecast" on the next, because the green "● LIVE" pill crowds the title bar. It reads as two disconnected labels.

Fix: shorten to "AI Forecast" or move the LIVE pill under the header.

UAT-08 **HIGH** "1685 Items" low-stock number is presented as an alarm without triage

That count is ~70% of the 2,429-item catalog. Presenting it in red with a bright "Email alert now" call-to-action guarantees alert fatigue and eventually a cashier hitting the button just to make it stop.

Fix: split into "Critical (0 stock)" vs "Below reorder point" and hide the email button behind a menu.

UAT-09 **HIGH** "GROWTH +32.5%" carries no timeframe

The tile shows a green pill "+32.5%" with no comparison window. Is that vs. yesterday? last month? forecast delta? The number under it (₱140,759 projected) is 30-day, so most users will read it as "we grew 32.5% this month" and be wrong.

Fix: label as "vs prior 30 days" or drop it.

UAT-10 **HIGH** Expiry Watchlist shows only "Expired" pills with no age

Every visible row (Whole Milk 1L, Sliced Bread, Cheddar Cheese) is tagged "Expired" but the user can't tell whether that means "expired yesterday" or "expired six months ago". The column reads "STATUS / DAYS" but the DAYS half is empty.

Fix: render "Expired 3d ago" / "Expires in 5d" using the anchor date the forecast already computes.

UAT-11 **MEDIUM** Sidebar label "Home" is ambiguous

In a POS/inventory app this reads as "back to landing page" rather than "the dashboard". Every peer product (Odoo, Loyverse, Square) calls it "Dashboard".

Fix: rename the sidebar item.

UAT-12 **MEDIUM** Notification bell has no unread badge

The bell in the top-right is a bare icon. Users can't tell there are unread alerts without clicking, which they won't.

Fix: add a red dot / numeric badge driven by the same feed that fills the bell popover.

UAT-13 **MEDIUM** Daily Sales Trend dots aren't connected — hard to read as a trend

The chart title says "Trend" but only scatter dots are drawn. A trend line or connected polyline would communicate direction.

Fix: switch Recharts <Scatter> to <Line>, add a 7-day rolling average behind it.

Screen 3 — Inventory Management

AMPC INVENTORY MANAGEMENT

Inventory List | **Supplier Reports**

PRODUCT LIST (2429 items)

PRODUCT	SUPPLIER	CATEGORY	STOCK	MIN STOCK	UNIT COST	PRICE
Whole Milk 1L 4800001001	Alpha Distributing Co.	Dairy	40	10	₱80.00	₱95.00
Cheddar Cheese Block 250g 4800001002	Alpha Distributing Co.	Dairy	24	10	₱150.00	₱180.00
Sliced Bread (Whole Wheat) 4800001003	Global Goods Inc.	Bakery	27	10	₱60.00	₱75.00
Canned Tuna in Oil 180g 4800001004	Global Goods Inc.	Canned Goods	82	10	₱42.00	₱55.00
Instant Noodles (Chicken) 4800001005	Global Goods Inc.	Pantry	150	10	₱13.00	₱16.00
Paracetamol 500mg (Box of 100) 4800001006	Prime Wholesale Ltd.	Pharmacy	20	20	₱280.00	₱350.00

Figure 3a — Inventory list, 2,429 items after AMPC import.

- UAT -14

HIGH

Min Stock is an always-editable input on every visible row

Every product row renders a `<input value="10">` under "MIN STOCK". Any accidental keystroke overwrites the reorder threshold without a save button. There's no confirm, no undo.

Fix: render as a static number, tap-to-edit a pencil icon; require Enter to commit and Esc to cancel.
- UAT -15

HIGH

Stock column shows numbers styled as clickable links but they do nothing

"40", "24", "27", "82" are blue and bold like anchor tags. Clicking them does nothing. False affordance.

Fix: either wire them to open a stock-movement history drawer, or make them plain text.
- UAT -16

MEDIUM

No visible pagination — 2,429 rows render into an endless scroll

On the test machine this dropped the frame rate noticeably while scrolling. Cheap store hardware will struggle.

Fix: paginate at 50 rows or add react-window virtualization.
- UAT -17

MEDIUM

"Export Inventory Sheet" doesn't say what will be exported or in what format

Excel? CSV? Filtered? All products? A first-time click is a guess.

Fix: label as "Export .xlsx" and, when a search or category filter is active, subtitle "Export filtered (N items)".

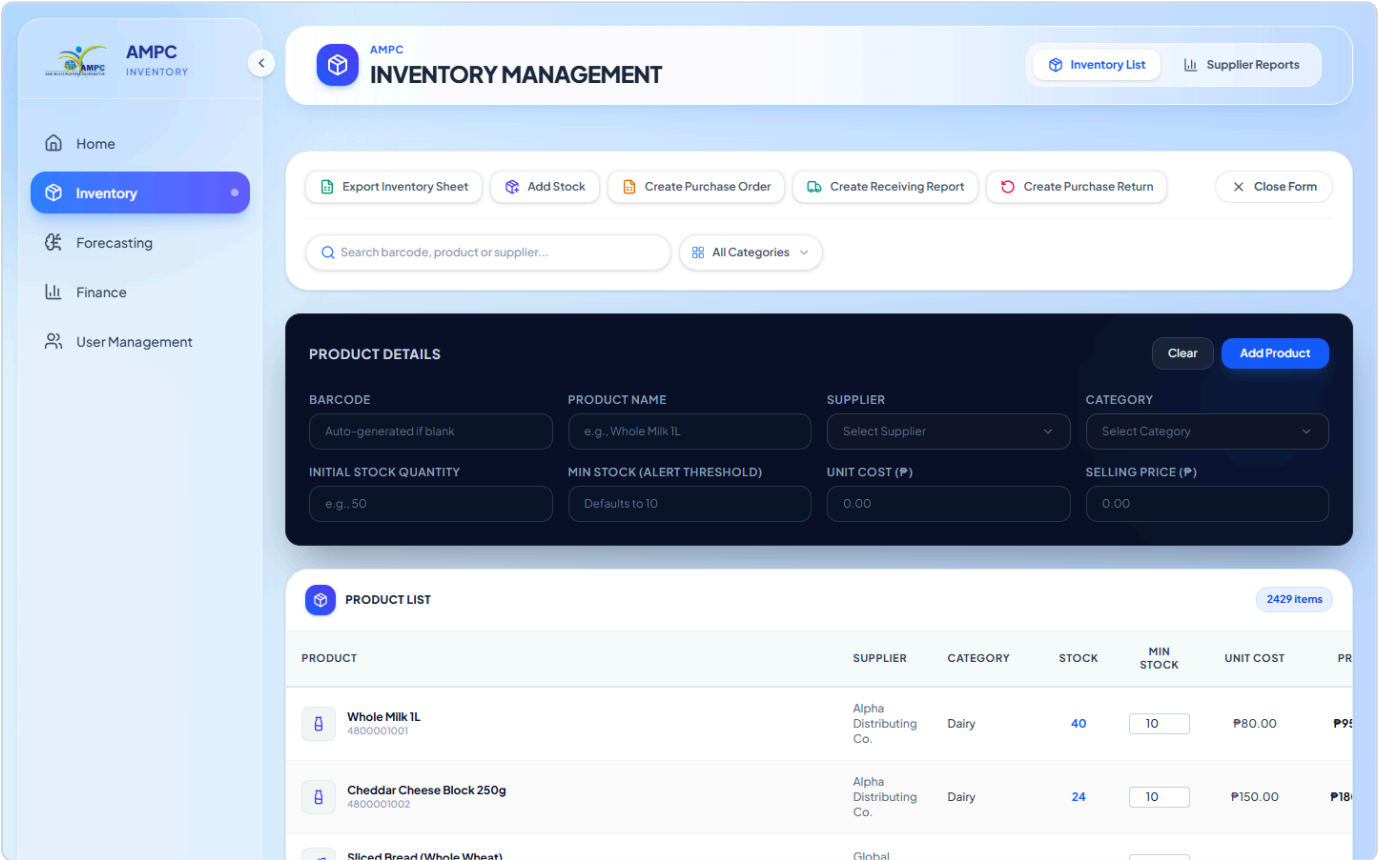


Figure 3b — Add Product opens as an inline panel, pushing the list down.

- UAT -18

MEDIUM

Add Product opens as an inline panel that pushes content out of view

If the admin scrolls the list to double-check a similar SKU, the form scrolls too and loses focus. It also visually competes with the list header.

Fix: turn it into a right-side slide-in drawer (matches the Purchase Order modal pattern already in the codebase).

UAT -19

MEDIUM

Category dropdown offers no way to add a new category on the fly

After the AMPC import there are 30+ categories in use. If the admin needs a 31st, they can't create it here — they have to leave the form.

Fix: combobox that accepts free text and offers "Create 'new value'" at the bottom of the suggestion list.

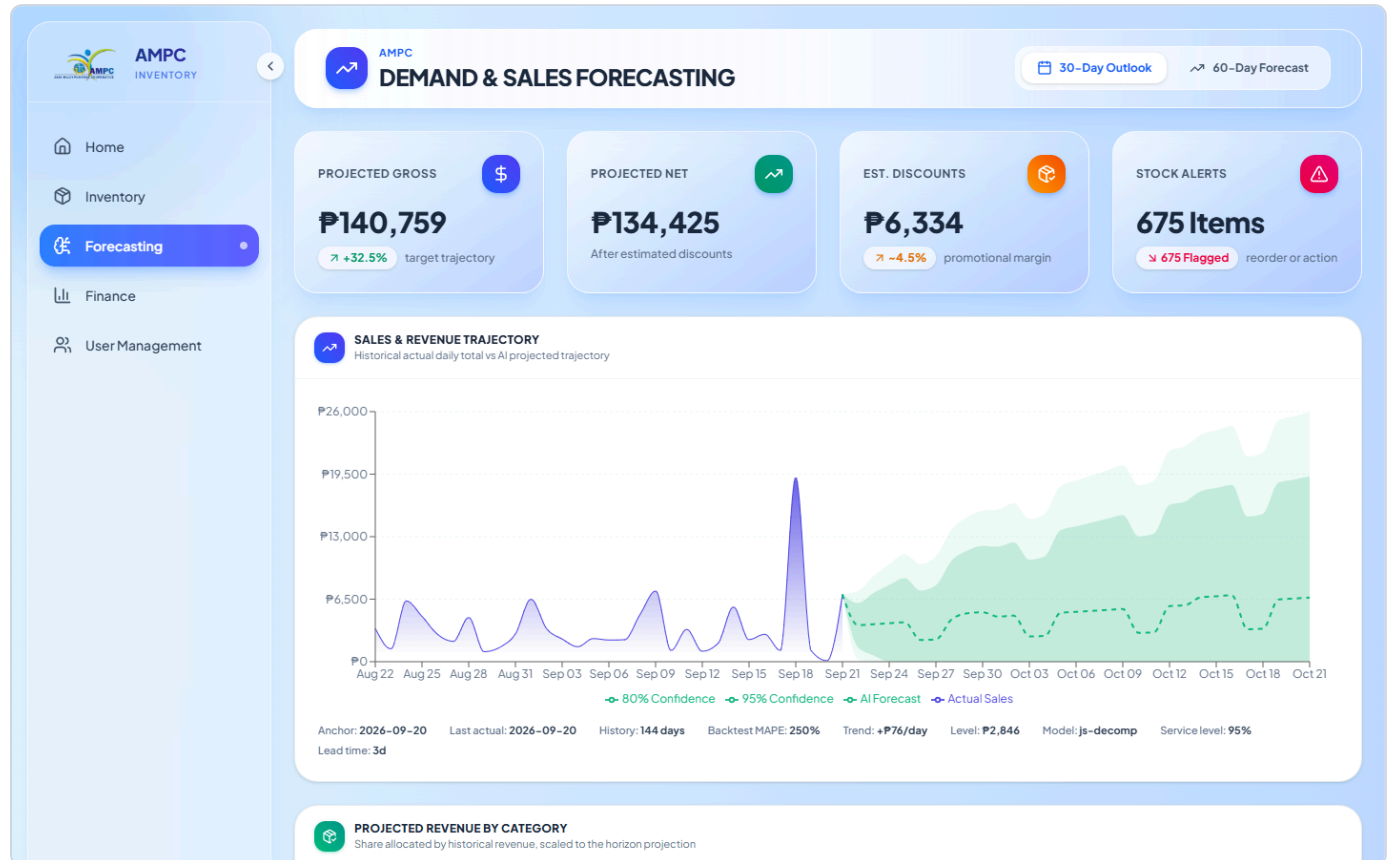
Screen 4 — Demand & Sales Forecasting

Figure 4a — 30-day outlook.

UAT -20

HIGH

Backtest MAPE 250% is presented as a KPI, not a health warning

A MAPE of 250% means the point forecast is off by 2.5× on average. To a coop administrator, "Backtest MAPE: 250%" reads like "look how much data we have" — not "the model is unreliable this month".

Fix: colour-code the MAPE (green < 20%, amber < 50%, red otherwise) and add a "?" tooltip: "average absolute error of the last 7-day backtest".

UAT -21

MEDIUM

Chart legend entries are visually indistinguishable

All four legend entries (80% Confidence, 95% Confidence, AI Forecast, Actual Sales) use identical dashed round markers. Colour alone is doing the work and the bands are close in hue.

Fix: use solid swatches for the bands and dashed line for the forecast; darken the actual-sales fill.

UAT -22

MEDIUM

Diagnostics footer is 11px gray — nearly unreadable

The strip "Anchor / Last actual / History / Backtest MAPE / Trend / Level / Model / Service level / Lead time" packs eight critical facts in tiny type on a light background. A supervisor giving a demo can't point at it.

Fix: bump to 12.5px, darker slate; break into two rows if it wraps.

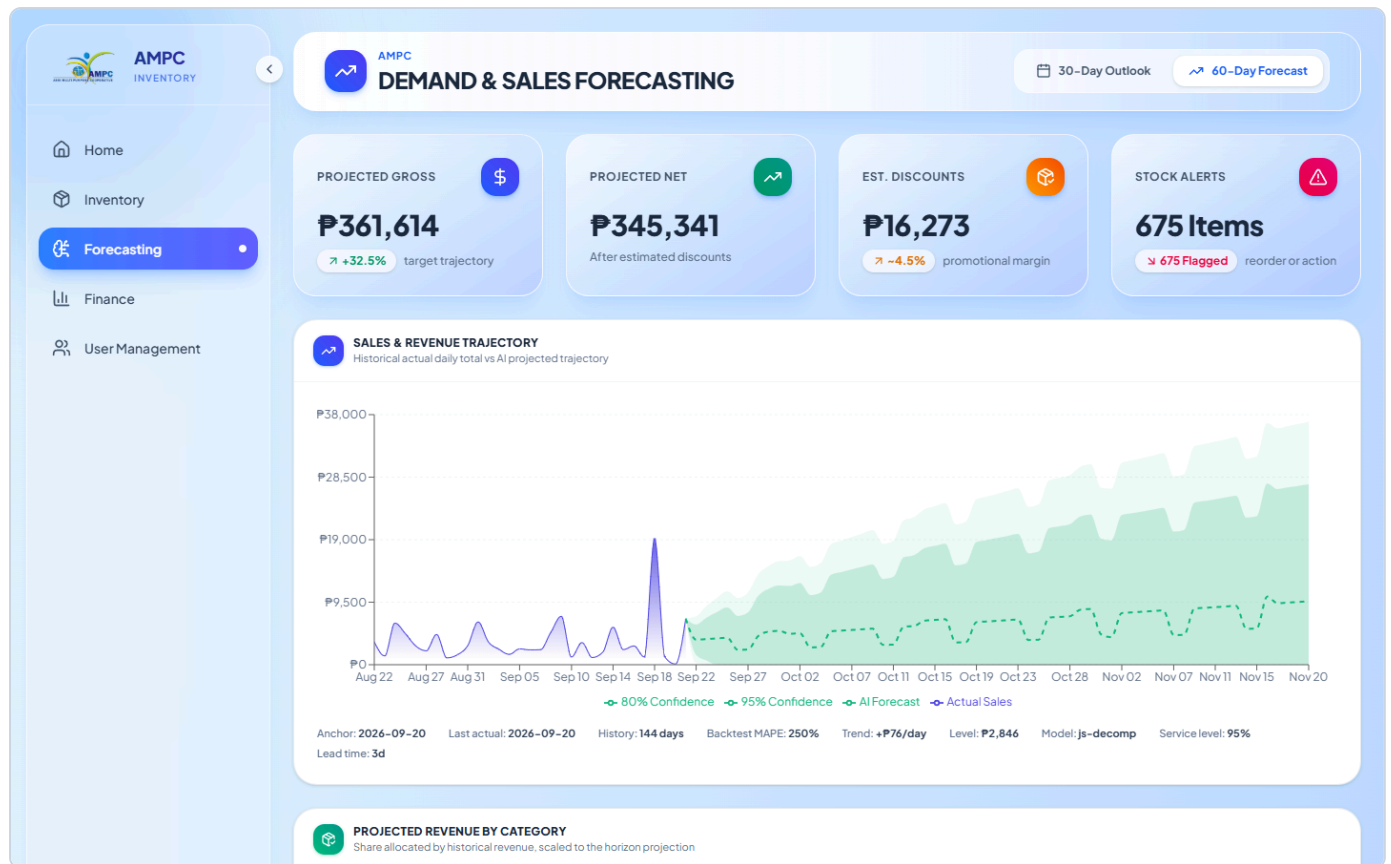


Figure 4b — 60-day forecast: prediction bands fan out visibly, useful.

UAT-23

MEDIUM

Toggle 30- ↔ 60-day updates numbers but has no loading indicator

The KPI tiles change silently after ~600ms. On slower hardware or a bad Wi-Fi, the user won't know something is loading.

Fix: skeleton state on the KPI cards during the fetch.

Screen 5 — Finance & Audit Control

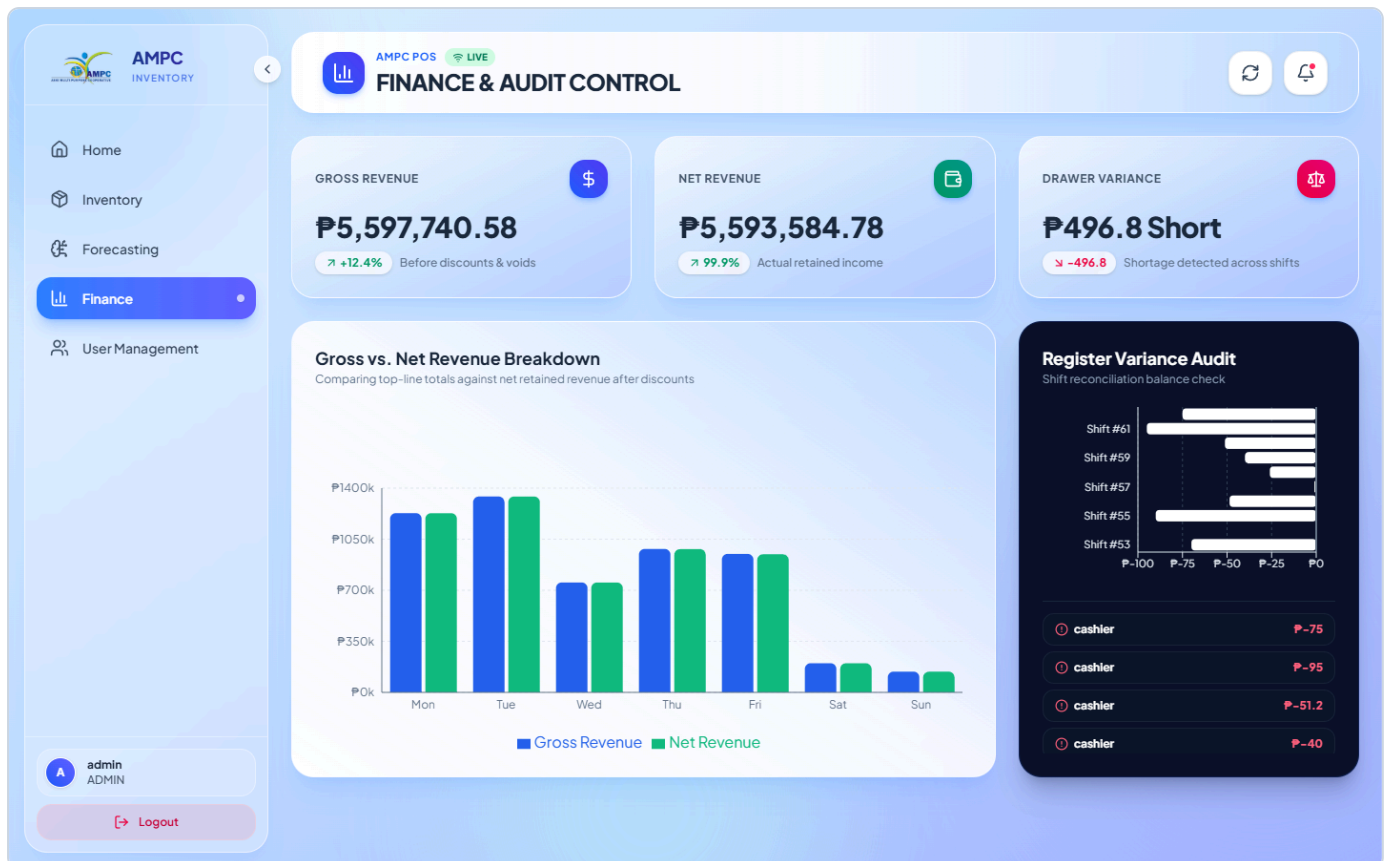


Figure 5 — Finance overview after AMPC import (₱5.5M+ gross).

UAT-24 HIGH "NET REVENUE 99.9%" green pill is misleading

A 99.9% figure next to Net Revenue reads as "profit margin". It's actually "net / gross" — a truism if discounts are near zero. Admins will confuse this with margin.

Fix: label as "Retained after discounts" or drop the pill.

UAT-25 HIGH Register Variance Audit rows all say "cashier" — no shift/date to distinguish

Four rows show "cashier ₱-75", "cashier ₱-95", "cashier ₱-51.2", "cashier ₱-40". Which shift? Which date? Without that, an admin can't investigate the shortage.

Fix: show shift #61 · cashier · Sep 21 AM per row.

UAT-26 MEDIUM Two top-right icons (refresh, bell) with no tooltips

Both are plain icons with no labels. First-time users have to click to guess.

Fix: title="" tooltip on hover, or use a Lucide + text pattern like elsewhere.

UAT-27 MEDIUM Weekday Gross vs Net chart has no y-axis density; bars for Sat/Sun collapse

Weekend bars are tiny relative to Tue/Thu. A log-scale toggle or a "% of weekday max" secondary line would help spot slow days.

Fix: add a "% of best day" caption per bar.

Screen 6 — User Management (protected)

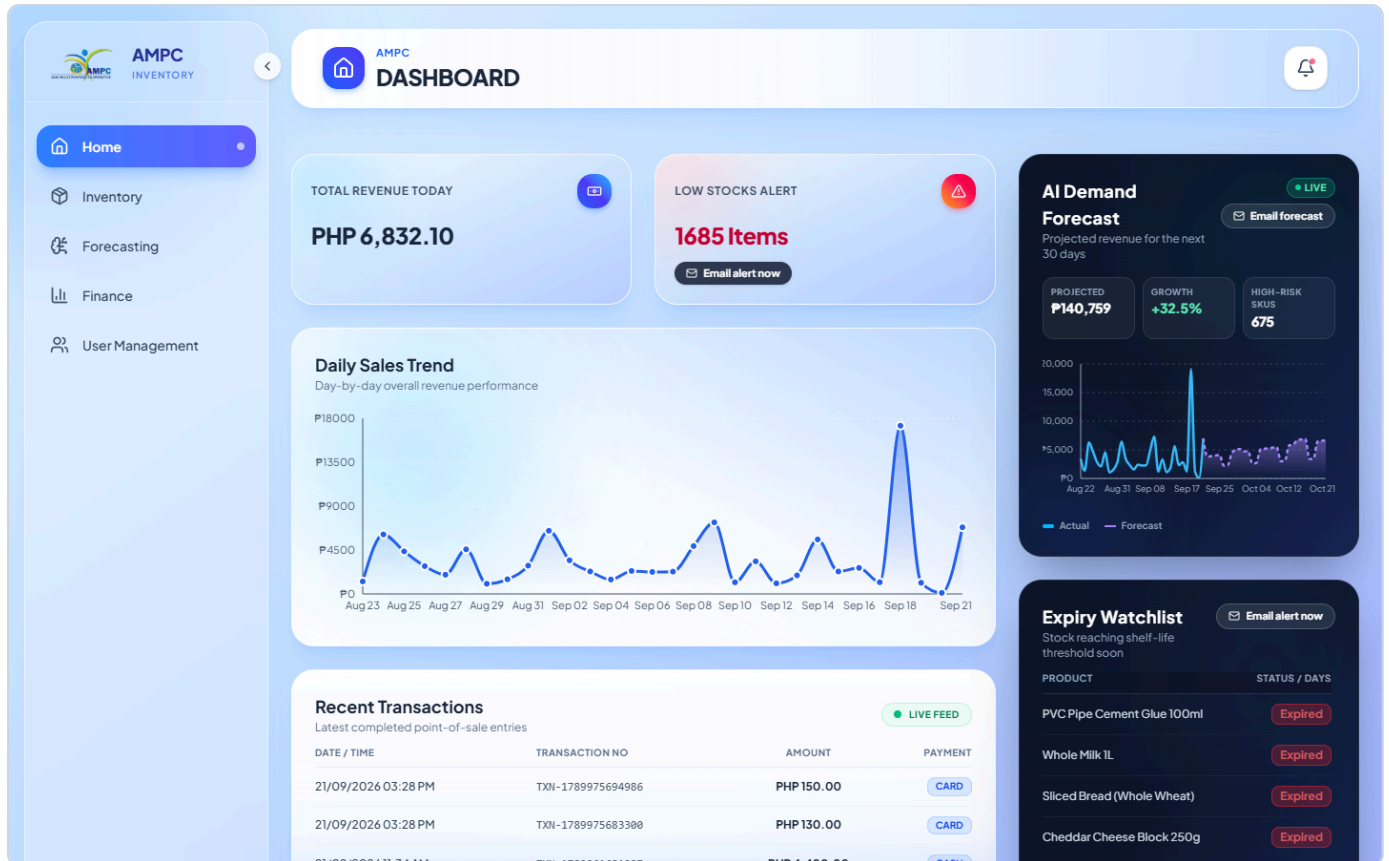


Figure 6 — Navigating directly to `/pages/ims/UserManagement` silently drops the user back on the Dashboard.

UAT-28 **CRITICAL** Direct URL to protected route silently redirects with no explanation

Deep-linking to `/pages/ims/UserManagement` lands on the Dashboard. The user (or a bookmark, or a support link) has no idea their request was rejected. If the sidebar guard is the canonical entry, the route itself should either open the admin-auth modal automatically or render an inline "Verify to continue" state.

Fix: move the `isAdminAuthenticated` guard inside the route component so the modal fires on direct navigation too.

UAT-29 **MEDIUM** Double auth: role check + password re-verify

The sidebar only shows "User Management" to ADMIN, then still demands the admin password before opening. Either the role check is enough, or the password should replace it — asking for both is friction without extra safety in a single-session context.

Fix: keep the modal only for destructive actions (delete/deactivate), not for the read-only landing.

Screen 7 — Cashier POS

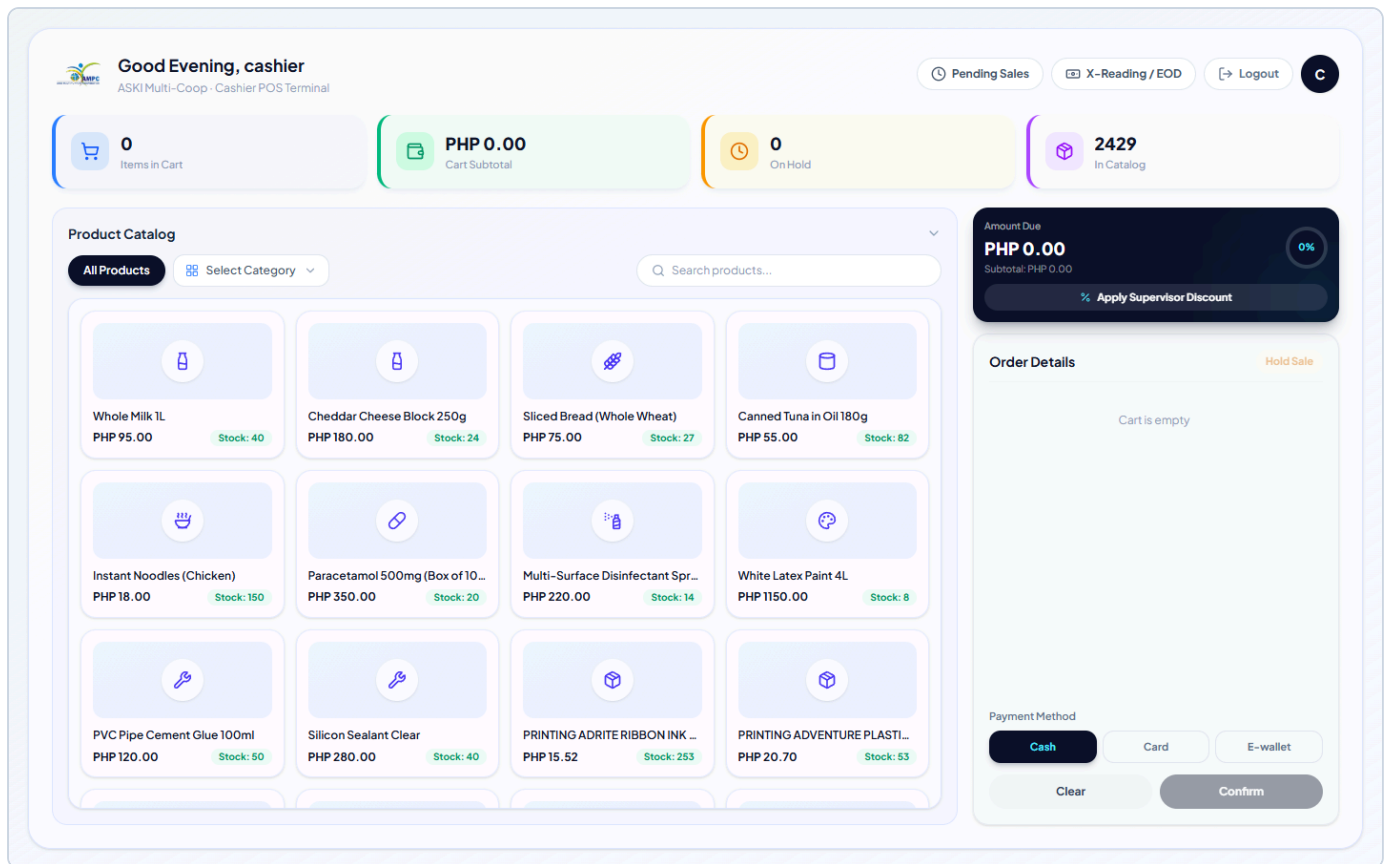


Figure 7a — POS opened as cashier, empty cart.

UAT -30 **HIGH** Every product tile shares one placeholder icon

2,429 SKUs, one graphic. Cashiers use visual grouping to spot the right SKU; identical placeholders force them to read every title. Real-world POS operators lose seconds per line item.

Fix: even a category-icon fallback (bottle, snack, printer, etc.) would help; a real image URL on the Product schema is the long-term answer.

UAT -31 **HIGH** Product name truncation with no tooltip

"PRINTING ADRITE RIBBON INK ..." and "Multi-Surface Disinfectant Spr..." cut off silently. Hovering does nothing. Cashiers can't verify SKU before adding.

Fix: native title on the truncated span, or a hover popover.

UAT -32 **HIGH** Logout is a single click away from a live shift

The top bar shows Pending Sales · X-Reading/EOD · **Logout** · avatar. One misclick during a busy queue ends the shift and voids the cart on the next re-login. In real coop cashier flow this is a repeated fear.

Fix: move Logout behind the avatar menu with a "You have items in the cart. Log out anyway?" confirmation.

UAT -33 **HIGH** "Confirm" button greyed with no reason label

First-time cashiers click it, nothing happens, they click harder. No tooltip, no "Add products to enable".

Fix: replace the disabled state with a live label ("Cart is empty" / "Select payment method").

UAT -34 **MEDIUM** Category dropdown label lies about state

"All Products" toggle is active but the adjacent dropdown still says "Select Category". Users read them as separate contradictory filters.

Fix: when "All Products" is active, dim the dropdown and set its label to "Any".

UAT-35

MEDIUM

Amount Due shows a 0% pill next to ₱0.00 — meaningless in empty state

The right-side panel shows "Amount Due ₱0.00 0%" with a subtotal ₱0.00 underneath. The 0% is meant to reflect discount; on an empty cart it's noise.

Fix: hide the 0% pill until the cart has ≥ 1 item.

UAT-36

LOW

"Cart is empty" state has no next-step guidance

Just the sentence, centred. No icon, no "Scan a barcode or tap a product to start."

Fix: illustrated empty state with two hints — scanner icon + product-tile icon.

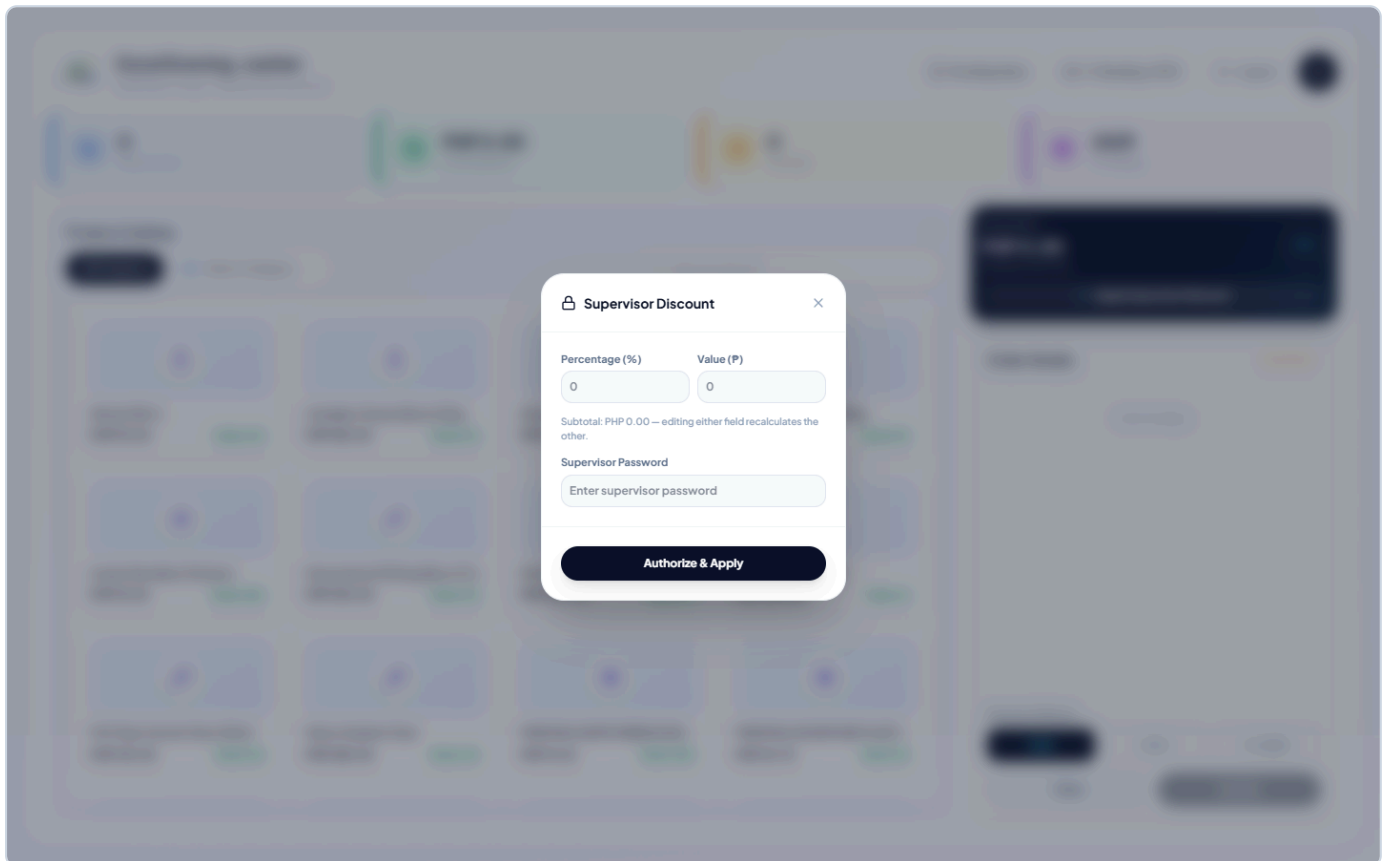


Figure 7b — Supervisor Discount modal.

UAT-37

HIGH

Supervisor Discount modal opens with an empty cart

Clicking "Apply Supervisor Discount" with no items in the cart still opens the modal. Typing anything ends up producing a "cannot apply" error only after Authorize & Apply.

Fix: disable the trigger button until `cart.length > 0`.

UAT-38

HIGH

Supervisor Password field has no show/hide toggle

Same fix as UAT-02 — but especially painful here because the supervisor is standing behind the cashier at rush hour.

Fix: add the eye toggle used in the Admin modal.

UAT-39

MEDIUM

"Percentage (%)" and "Value (P)" both default to 0 — no visual "either or"

The subtitle "editing either field recalculates the other" is buried in small type. Cashiers will type both by reflex and be confused when one overwrites the other.

Fix: radio-style toggle "Percent | Peso" that hides the inactive field.

Supervisor Discount



Percentage (%)

0

Value (P)

0

Subtotal: PHP 0.00 — editing either field recalculates the other.

Supervisor Password

Enter supervisor password

Authorize & Apply

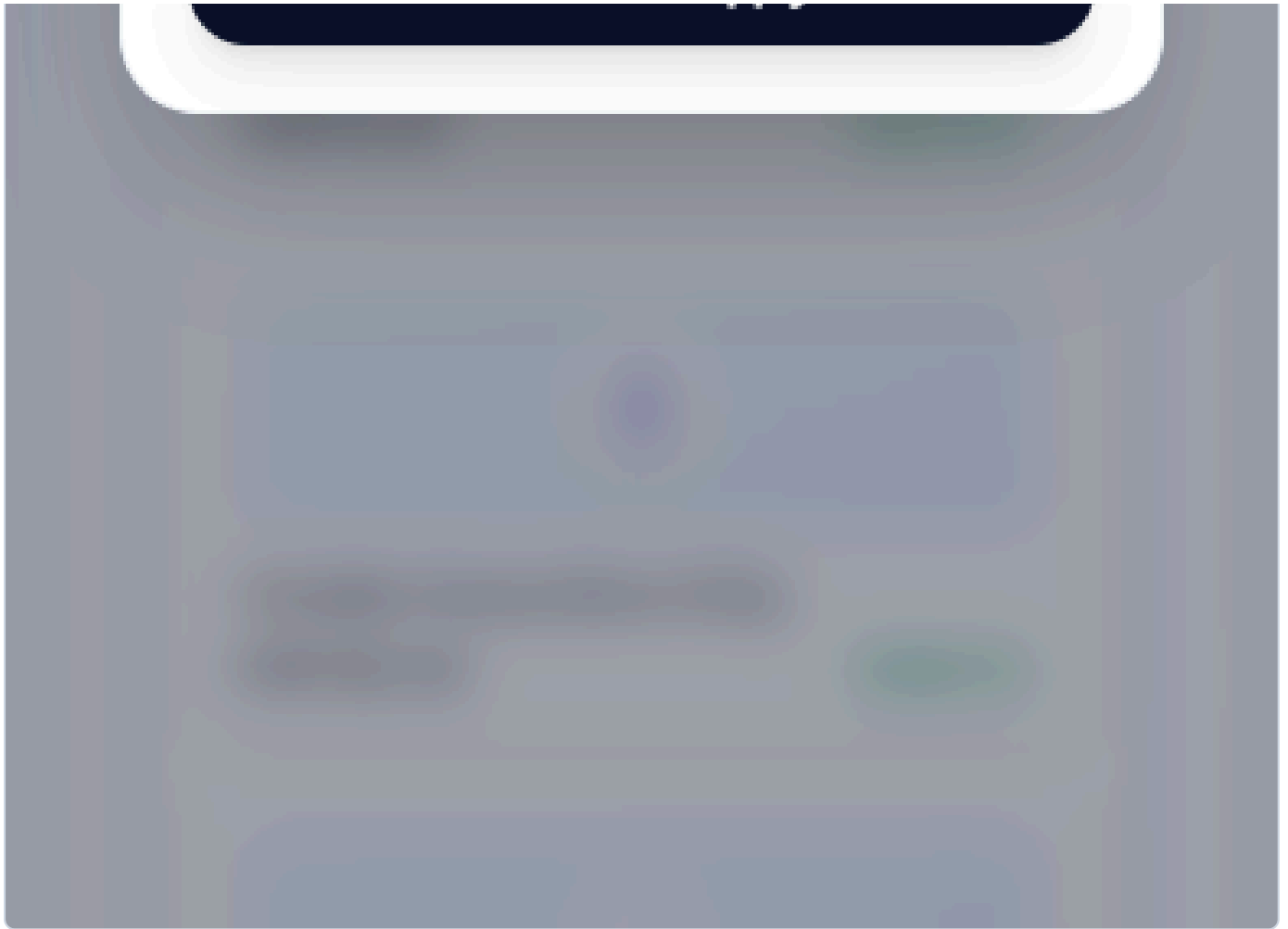


Figure 8a — Discount modal on a phone — the modal itself is fine but the background POS UI is unusable at this width.

UAT-40

HIGH

No responsive layout — sidebar never collapses on mobile

At 390px the sidebar shrinks but there is no hamburger toggle to reclaim horizontal space, POS cards squeeze into unreadable widths, and the dashboard KPIs vanish off-screen.

Fix: hide the persistent sidebar under md; expose a top hamburger; make the POS product grid switch from 4-col to 2-col below 768px.



AMPC Inventory

Username

Enter your username

Password

Enter your password

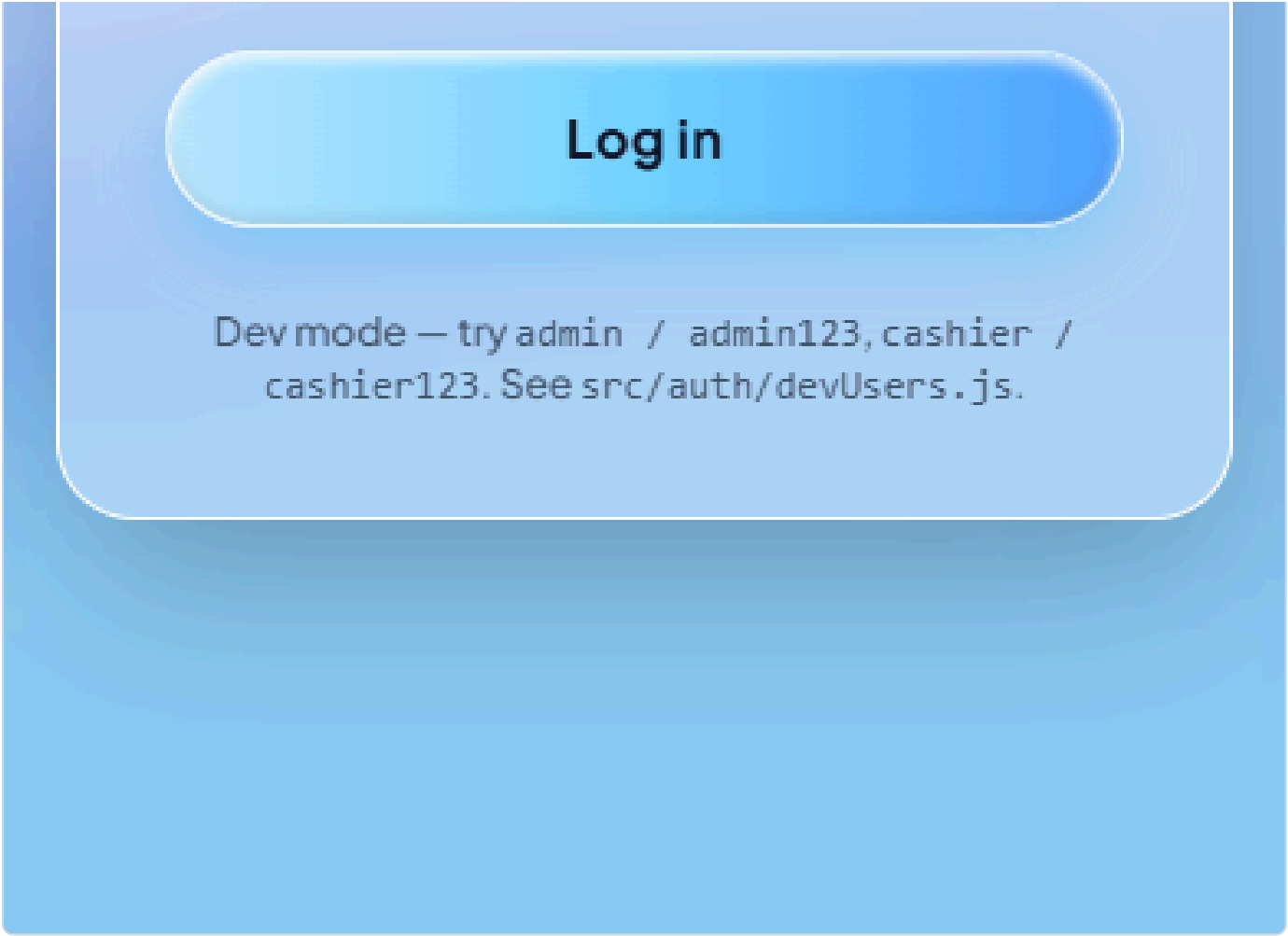


Figure 8b — Same session on mobile: because sessionStorage was cleared, the user is back on login. The login card itself scales cleanly.

Prioritised backlog

ID	Sev	Where	What to change
UAT-01	Critical	Login	Gate dev-mode credential hint behind <code>import.meta.env.DEV</code>
UAT-05	Critical	Global	Unify currency to ₱ via <code>formatPeso()</code>
UAT-28	Critical	User Mgmt route	Trigger admin-auth modal on the route itself, not only in sidebar
UAT-02	High	Login	Password show/hide toggle
UAT-06	High	Dashboard	Add date subtitle to "Total Revenue Today"
UAT-07	High	Dashboard	Fix AI Forecast tile title wrap
UAT-08	High	Dashboard	Split "Low Stocks Alert" into critical vs below-reorder
UAT-09	High	Dashboard	Label growth timeframe
UAT-10	High	Dashboard	Fill "DAYS" column on Expiry Watchlist
UAT-14	High	Inventory	Tap-to-edit Min Stock, not always-editable
UAT-15	High	Inventory	Either wire Stock number to a drawer, or de-style it
UAT-20	High	Forecasting	Colour-code MAPE + tooltip
UAT-24	High	Finance	Remove/relabel the "99.9%" NET Revenue pill
UAT-25	High	Finance	Include shift/date per variance row

UAT-30	High	POS	Category-icon fallback for product tiles
UAT-31	High	POS	Tooltip on truncated product names
UAT-32	High	POS	Move Logout behind avatar; confirm on cart-not-empty
UAT-33	High	POS	Replace disabled Confirm with reason label
UAT-37	High	POS	Disable discount trigger on empty cart
UAT-38	High	POS	Show/hide toggle on supervisor password
UAT-40	High	Mobile	Sidebar → hamburger under 768px; 2-col product grid

Notes for QA (not counted above)

- Puppeteer script backend/scripts/uatCapture.js reproduces this walk-through in ~90s. Run it after every UI PR to spot regressions.
- Forecast Python microservice was unreachable during this walk-through (pandas==3.0.6 pinned in ai-service/requirements.txt does not exist on PyPI) — the JS fallback is what all screenshots depict. Fixing the pin is a separate follow-up.
- The Aug/Sep transactions visible on charts come from prisma/seeder.js, not from the AMPC workbook — see conversation history.

Generated 2026-09-21 from backend/scripts/uatCapture.js. Screenshot evidence in docs/uat/screens/.